

are installed on a computer to allow complete control and great security, as Bitcoin is accessed from specific users' computers. The software called Bitcoin Core was developed and supported by the Bitcoin Foundation. It creates a *wallet.dat* file which stores the data with regard to a user's personal Bitcoin wallet. It is of three types — desktop, mobile and online.

Desktop wallets: These are downloaded to a computer and accessed from that device only. They are secure, but whenever there is any computer problem, the wallet software gets impacted. Examples are the Exodus and Electrum wallets.

Mobile wallets: These are wallets that run only on smartphones using the app, and provide portability and flexibility to the user. Examples include Jaxx, Freewallet and Infinito wallet.

Online wallets: These are wallets that run on the cloud and can be accessed from any device. An example is Guarda.

- **Paper wallets:** These are termed a 'software application'. Here the user has to print a QR code for both the public and private keys to send and receive digital currency. These are categorised as cold wallets as the keys are stored offline. An example is the Verge Paper wallet.
- **Miscellaneous wallets**
 - a. **Full node wallet:** In this case, the user controls private keys and hosts a full copy of the blockchain.
 - b. **Custodial wallet:** This type of wallet lets the user control private keys. Most exchange wallets are custodial wallets.
 - c. **Coin-specific:** Such wallets only work with specific coins.
 - d. **Network-specific:** These wallets hold multiple tokens on a single network.
 - e. **Universal/multi-coin wallet:** Such wallets hold addresses from multiple coins.

A guide to protecting cryptocurrency wallets

- **Regular backup:** All types of software wallets should be backed up at regular intervals of time, for security. In the case of desktop wallets, the *wallet.dat* file should be encrypted with a password and multiple copies should be made on other computers or pen drives. In the case of online backups, regular password changes with a strong password are recommended. It is also best to have multi-factor authentication, for security.
- **Regular updates:** All the wallets should be updated with the latest Bitcoin wallet software to guarantee the latest security enhancements.
- **Enhanced security layers:** The safety of your money is solely your responsibility, so try to add additional layers of security to your cryptocurrency wallet. Start with a strong password on all the devices you have the wallet software installed on. Choose wallet service providers with strong

security policies like two-factor authentication and pin code request every time a wallet application is opened.

The top open source cryptocurrency wallets

Cryptocurrency adaption is on rise across the world, so wallet selection becomes quite complex. In this article, some of the best and most secure open source cryptocurrency wallets are listed.

Breadwallet: Breadwallet is an open source cryptocurrency mobile wallet designed by the Bread company in 2015. The Breadwallet app directly links with the respective blockchains to increase the speed of transactions with a simple-to-use interface. Initially, Breadwallet was designed for Bitcoins only, but it has expanded to support other currencies also, like Ethereum, Bread Coin and Bitcoin Cash.

To use Breadwallet, the user has to follow these steps:

- Download the app from Google's Play Store or Apple's App Store on a smartphone.
- Navigate and click on 'Create New Wallet'.
- Select '6-Pin Code (Private Key)' to log into the wallet in order to run transactions.
- Use the code to log into the wallet. At this point, the wallet will generate a 'seed phrase' that will be required to recover the tokens in case the phone gets lost.
- To add tokens such as Bitcoin, you need to tap the 'Receive' icon. It will generate an address referred to as a QR code that you should use to direct the coins from the Bitcoin platform.
- To send coins, navigate to the 'Send' icon and enter the receiving address. Note that the app provides the option of scanning or writing the receiving address for easier management.

Features:

- Compatible with Android and iOS.
- There are no fees for performing transactions except the network fees of the coin in which the user is performing transactions.
- There is no need to sign up or register to perform transactions. It is simple and easy to use as it is an open source platform.

Pros: User friendly, free, open source, direct connection with Bitcoin blockchain, tough-ID, and the ability to buy Bitcoin directly using Bread App. Multiple wallet addresses are generated automatically for every transaction.

Cons: Frequent app crashes, lacks two-factor authentication, slow network synchronisation, lacks multi-signature transactions, limited altcoin support, and higher charges for transactions as compared to other wallets.

Official website: <https://brd.com/>